

Fynd ERP

Product Cookbook

A visual, self-serve guide to setting up Fynd ERP, launching production, operating the shop floor, and tracing every finished unit.

48	10
GUIDED ARTICLES	ORDERED STAGES

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STAGE 01

Start Here

Learn the manufacturing basics, understand how Fynd ERP records connect, and see the complete journey before configuring anything.

5 guided articles

Fynd ERP in Plain Language

Understand what Fynd ERP does without needing a manufacturing background.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/start-here/forge-mes-in-plain-language>

How it works

01	Plan what to make
02	Prepare the factory
03	Release production
04	Build each unit
05	Inspect and trace

What to do

- 1 Use Fynd ERP to turn a production plan into controlled work on real factory lines.
- 2 Give every unit a route, a production task, and a history that can be traced later.
- 3 Use checks, repair paths, packaging, and dashboards to keep production reliable.

Rules to remember

- Fynd ERP manages manufacturing execution; it does not replace the physical factory.
- Master data describes what can happen. Transaction data records what actually happened.
- The setup sequence matters because downstream records depend on upstream records.

Ready when

- [] I can explain the difference between setup data and production activity.
- [] I understand that a finished unit moves through a routing one station at a time.
- [] I know that Fynd ERP keeps a traceable record for every released unit.

Manufacturing Basics for New Users

Learn the small set of factory concepts used throughout the cookbook.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/start-here/manufacturing-basics>

How it works

01	Material
02	Operation
03	Station
04	Finished unit
05	Shipment

What to do

- 1 A product is the item the factory intends to make.
- 2 A bill of materials lists the components required to make that product.
- 3 A routing lists the operations and stations the unit must pass through.
- 4 A work order authorises a quantity of that product to be produced.
- 5 A production task represents one finished unit moving through the route.

Rules to remember

- A Line is a production lane; a Station is a specific place where an operation happens.
- A component can be scanned by serial number or consumed by quantity.
- A failed operation can send the unit to repair, rework, hold, or scrap depending on configuration.

Ready when

- [] I can distinguish a product, component, BOM, routing, work order, and production task.
- [] I understand the difference between a Line and a Station.
- [] I know why traceability begins before production is released.

How Fynd ERP Data Connects

See the dependency chain that connects factory setup to executable production.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/start-here/how-forge-data-connects>

Before you begin

- Fynd ERP in Plain Language
- Manufacturing Basics for New Users

How it works

01	Site and Line
02	Station and Shift
03	Product and active BOM
04	Routing and Line assignment
05	Work Order
06	Production Tasks

What to do

- 1 Sites locate Stations and can also group Lines.
- 2 Products use BOMs to define material requirements.
- 3 Routings connect operations to Stations and require an active BOM.
- 4 Product-to-Line assignments identify where a product can start production.
- 5 Releasing a Work Order creates unit-level Production Tasks on eligible Lines.

Rules to remember

- A missing prerequisite blocks the next record even when the next screen is visible.
- Projects and Product Families are recommended organisational layers, not hard execution blockers.
- Production Orders are optional grouping records; Work Orders create executable production.

Ready when

- [] I can follow the dependency chain from Site to Production Task.
- [] I know which organisational records are recommended.
- [] I understand why a Production Order alone does not start production.

Complete Fynd ERP Setup Journey

Follow the recommended order for configuring Fynd ERP and releasing the first production run.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/start-here/complete-setup-journey>

Before you begin

- How Fynd ERP Data Connects

How it works

01	Model the factory
02	Organise products
03	Define materials
04	Design the route
05	Assign starting Lines
06	Create and release a Work Order
07	Run, inspect, package, and trace

What to do

- 1 Review the Plant and create the required Sites.
- 2 Create Lines, mark the correct initial Line as a Starting Line, and configure capacity.
- 3 Create operation Stations, dedicated Repair Stations, and Shifts.
- 4 Create recommended Projects and Product Families, followed by Components and Products.
- 5 Review the draft BOM created with the Product and activate it.
- 6 Build and activate a Routing, then assign the Product to eligible Lines.
- 7 Create a Work Order with at least one Shift, generate unit serials, validate readiness, and release.

Rules to remember

- Do not create Work Orders until the Product has both a Routing and an active Line assignment.
- Do not release production until the initial Line is marked as a Starting Line.
- Do not bypass readiness warnings without understanding their effect on later Stations.

Ready when

- [] Every setup stage has an owner in my implementation plan.
- [] Dependencies are complete before the first Work Order is created.
- [] A small trial Work Order will be used before full-volume production.

How to Use This Cookbook

Use the cookbook as a guided setup path, task reference, and troubleshooting tool.

RECOMMENDED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/start-here/how-to-use-this-cookbook>

How it works

01	Choose a journey
02	Check prerequisites
03	Follow the steps
04	Confirm readiness
05	Continue to the next guide

What to do

- 1 New users should follow the sections in the order shown in the navigation.
- 2 Experienced users can use global search from any page to find a record or task.
- 3 Use the Required, Recommended, and Conditional labels to understand priority.
- 4 Use the readiness checklist at the end of each guide before continuing.

Rules to remember

- Required means Fynd ERP blocks or cannot reliably complete the downstream action without it.
- Recommended means Fynd ERP may allow the action, but the setup improves organisation and control.
- Conditional means the setup is needed only when that manufacturing capability is used.

Ready when

- [] I know how to search from every cookbook page.
- [] I understand the three requirement labels.
- [] I know where to download the complete cookbook as a PDF.

STAGE 02

Configure Your Factory

Create the physical structure Fynd ERP needs: Sites, Lines, Stations, Repair Stations, capacity, and Shifts.

6 guided articles

The Fynd ERP Factory Model

Understand how Plant, Site, Line, and Station represent a real manufacturing unit.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/configure-factory/factory-model>

Before you begin

- Complete Fynd ERP Setup Journey

How it works

01	Plant
02	Site
03	Line
04	Station
05	Operation

What to do

- 1 The Plant is the overall manufacturing context used to isolate records.
- 2 A Site represents a physical location within the Plant.
- 3 A Line represents a production lane and receives Production Tasks.
- 4 A Station represents a specific operational point used in a Routing.

Rules to remember

- Stations require a Site.
- Lines can reference a Site, but the current Line setup does not require it.
- A Station can exist without a Line assignment, although linking the physical structure is recommended.

Ready when

- [] Every operation has a clear physical Site.
- [] Production lanes are represented as Lines.
- [] Work areas are represented as Stations rather than additional Lines.

Create Sites

Create the physical locations required before Stations can be configured.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/configure-factory/sites>

Before you begin

- The Fynd ERP Factory Model

How it works

01	Create Site
02	Required identity
03	Location details
04	Optional metadata
05	Use Site on Stations

What to do

- 1 Create one Site for each physical manufacturing location that needs separate operational records.
- 2 Give each Site a unique name and code within the Plant.
- 3 Add optional address details by selecting Country before State, and State before City.
- 4 Keep optional metadata within the 32 KB limit.
- 5 Use the Site when creating Stations and, where useful, Lines and storage locations.

Rules to remember

- Site name and code are the required fields; the code accepts uppercase letters, numbers, and hyphens only.
- A Station cannot be created without a valid Site.
- Site names and codes must be unique within the Plant.
- Do not create a separate Site for every room or workstation; use Stations for that detail.

Ready when

- [] Site names and codes are unique.
- [] Every planned Station can be mapped to a Site.
- [] Users can distinguish Sites without relying only on codes.

Create Production Lines

Create the production lanes that will receive and balance unit-level Production Tasks.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/configure-factory/lines>

Before you begin

- The Fynd ERP Factory Model

How it works

01	Create Line
02	Choose Line type
03	Set capacity
04	Mark Starting Line
05	Assign Product

What to do

- 1 Create each real production lane with a unique name and code.
- 2 Choose the Line type that best describes the manufacturing phase.
- 3 Set Capacity Per Shift above Capacity Per Hour and enter a whole-number Task Queue Capacity from 1 to 100,000.
- 4 Mark the Line that receives newly released units as a Starting Line.
- 5 Keep the Line active when it should be eligible for production.

Rules to remember

- Name, code, and Line type are required; the Line code has a maximum length of 50 characters.
- A Product needs at least one active Line assignment before a Work Order can be created.
- Normal initial release uses only eligible Lines marked as Starting Lines.
- Use a downstream phase as a Starting Line only when production can genuinely begin there.

Ready when

- [] Every production lane has a unique Line.
- [] The initial Line is marked as a Starting Line.
- [] Capacity reflects the number of active and waiting tasks the Line can accept.

Create Stations and Repair Stations

Create operational work points and the separate repair destinations required by Routings.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/configure-factory/stations-and-repair-stations>

Before you begin

- Create Sites

How it works

01	Create operation Station
02	Required identity
03	Validate identity
04	Create Repair Station
05	Use both in Routing

What to do

- 1 Create a Station for each operation that must be represented in a Routing.
- 2 Enter the required name, code, Site, and Station Category.
- 3 Use a name with at least two characters and a code containing only letters, numbers, and hyphens.
- 4 Create dedicated Repair-type Stations for failure handling.
- 5 Use a Repair Station different from the main operation Station in every Routing operation.

Rules to remember

- Station name, code, Site, and Station Category are required.
- Station names and codes must be unique within the Plant.
- Every operation node in the current Routing builder requires a Repair Station.
- The Repair Station cannot be the same Station as the operation itself.
- Automatic Stations are used for configured automation; regular Stations expect operator interaction.

Ready when

- [] Every planned operation has a Station.
- [] Every operation can point to a separate Repair Station.
- [] Station types match the way work will be performed.

Create Shifts

Define when production runs and make Shifts available for Work Order assignment.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/configure-factory/shifts>

Before you begin

- The Fynd ERP Factory Model

How it works

01	Define Shift
02	Required schedule
03	Valid times
04	Breaks
05	Schedule
06	Assign to Work Order

What to do

- 1 Create Shift definitions that reflect the Plant's working periods.
- 2 Enter the required name, unique code, Shift type, start time, and end time.
- 3 Use 24-hour HH:mm times; overnight Shifts are supported.
- 4 Give every break a name, type, and duration of at least one minute.
- 5 Add schedule information when production varies by day or calendar.
- 6 Select at least one valid Shift when creating each Work Order.

Rules to remember

- Shift codes use uppercase letters, numbers, and hyphens.
- Unpaid breaks are deducted from net working time.
- A Work Order cannot be created or started without at least one Shift assignment.
- Assigned Shifts must belong to the same Plant.

Ready when

- [] All active production periods are represented.
- [] Shift times do not overlap unintentionally.
- [] At least one Shift is ready for the first Work Order.

Factory Readiness Checklist

Confirm the physical factory model is complete before creating production master data.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/configure-factory/factory-readiness>

Before you begin

- Create Sites
- Create Production Lines
- Create Stations and Repair Stations
- Create Shifts

How it works

01	Sites ready
02	Lines ready
03	Stations ready
04	Shifts ready
05	Factory model approved

What to do

- 1 Walk through the physical production process and match each area to a Fynd ERP record.
- 2 Confirm which Line receives the first operation for every product type.
- 3 Confirm Line capacities using a small realistic production quantity.
- 4 Confirm every operation and failure path has a usable Station.

Rules to remember

- Do not compensate for missing Lines by creating extra Stations.
- Do not mark every Line as a Starting Line unless units can genuinely begin there.
- Repair Stations must be planned before Routings are built.

Ready when

- [] Required Sites exist.
- [] Lines are active, correctly typed, and have explicit capacity.
- [] Initial Lines are marked as Starting Lines.
- [] Operation and Repair Stations exist at valid Sites.
- [] At least one valid Shift exists.

STAGE 03

Define Products and Materials

Organise products, create components and variants, build BOMs, and choose the correct material tracking approach.

6 guided articles

Projects and Product Families

Use the recommended hierarchy to keep products, reporting, and future growth organised.

RECOMMENDED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/define-products/projects-and-product-families>

Before you begin

- Factory Readiness Checklist

How it works

01	Project
02	Product Family
03	Product
04	Variant

What to do

- 1 Create a Project for a meaningful programme, product programme, or managed manufacturing scope.
- 2 Create Product Families inside Projects to group closely related Products.
- 3 Assign Products to the appropriate Product Family during Product setup.
- 4 Use consistent naming so Work Orders and reports can inherit a useful hierarchy.

Rules to remember

- A Product Family requires a Project.
- Fynd ERP can create a Product without a Product Family, but the hierarchy improves organisation and reporting.
- Use Variants for versions of the same Product, not for unrelated Products.

Ready when

- [] Projects have a clear purpose.
- [] Product Families group genuinely related Products.
- [] The hierarchy will remain understandable as more Products are added.

Create and Update Materials in Item Master

Create material and part records in Item Master, update allowed details, and open their inventory tracking history.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/define-products/components>

Before you begin

- Projects and Product Families

How it works

01	Define component
02	Required identity
03	Choose tracking mode
04	Fixed Serial value
05	Stock thresholds
06	Add inventory

What to do

- 1 Create a Component for every material or part that must be planned, scanned, consumed, or traced.
- 2 Enter the required name, code, and unit; names and codes require at least two characters.
- 3 Choose one mutually exclusive mode: Serialized, Consumable, or Fixed Serial.
- 4 For Fixed Serial, enter the permanent barcode or QR value.
- 5 Set maximum stock above minimum stock and keep the reorder level between them.
- 6 Load or receive inventory before the Component is needed for production.

Rules to remember

- Unit and tracking settings cannot be changed after creation.
- BOMs reference Component records, so Components must exist first.
- Serialised Components require a valid individual identifier during consumption.
- Quantity-based Components are validated against the required BOM quantity.

Ready when

- [] Every BOM material has a Component record.
- [] Tracking type matches the physical identification method.
- [] Units of measure and quantities are consistent.

Create Products and Variants

Create finished-product definitions and decide when Variants are useful.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/define-products/products-and-variants>

Before you begin

- Create and Update Materials in Item Master

How it works

01	Create Product
02	Choose Product Family
03	Add optional Variants
04	Cover every Variant
05	Apply overrides
06	Configure serial format

What to do

- 1 Create the Product with a unique name of at least two characters and SKU of at least three characters.
- 2 Assign the recommended Product Family when available.
- 3 Add at least two Variants when the same Product has meaningful configurations.
- 4 Add a shared BOM or provide BOM entries for every Variant.
- 5 Use shared BOM rows for every Variant and Variant rows only to override matching quantities.
- 6 Configure a serial number template when a specific finished-unit format is needed.

Rules to remember

- A Product without Variants requires a shared BOM.
- A Product with Variants needs a shared BOM or a BOM for every Variant.
- If Variants are used, a Work Order must select the required Variant.
- The serial number template is optional, but unit serial generation before release is not.

Ready when

- [] Product name and SKU are unique.
- [] Variant use is deliberate and consistent.
- [] Every Product configuration has BOM coverage.

Build the Bill of Materials

Define exactly which Components and quantities are required to build the Product.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/define-products/bill-of-materials>

Before you begin

- Create Products and Variants

How it works

01	Select Product
02	Add Components
03	Set quantities
04	Review Variant overrides
05	Approve and activate

What to do

- 1 Review the initial draft BOM created during Product setup.
- 2 Confirm every required Component and quantity per finished unit.
- 3 Use shared entries for common materials and Variant entries only where requirements differ.
- 4 Review units of measure and effective quantities before activation.
- 5 Resolve non-positive quantities and duplicate Components within the same scope.

Rules to remember

- BOM items can be edited only while the Version is Draft.
- Routing creation requires an active BOM belonging to the selected Product.
- The pre-start readiness check uses the active BOM to calculate material requirements.
- Changing an active BOM can change future production readiness and should be controlled.

Ready when

- [] The BOM contains every required production Component.
- [] Quantities match one finished unit.
- [] Variant overrides do not duplicate shared requirements incorrectly.

Activate a BOM Version

Move the reviewed BOM from draft to the active production version required by Routing.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/define-products/activate-bom>

Before you begin

- Build the Bill of Materials

How it works

01	Open draft BOM
02	Review items
03	Release and lock
04	Verify Active status

What to do

- 1 Open the Product's BOM Version and review its status and items.
- 2 Confirm the BOM contains at least one item.
- 3 Select Release BOM Version and confirm the edit-lock warning.
- 4 Verify the active status, removed editing controls, and enabled Create Routing action.

Rules to remember

- Only reviewed draft or approved versions should be released.
- An empty BOM cannot be released as a meaningful production definition.
- Releasing locks the BOM Version, and only one Version is active for the Product at a time.
- Routing must use an active BOM from the same Product.

Ready when

- [] The correct BOM Version is active.
- [] The active BOM belongs to the intended Product.
- [] Material inventory exists or has a replenishment plan.

Track Serial, Lot, and Mixed Materials in Item Master

Receive and trace Item Master materials by lot, serial, quantity, or a mixed tracking model.

CONDITIONAL

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/define-products/material-tracking>

Before you begin

- Create and Update Materials in Item Master
- Build the Bill of Materials

How it works

01	Lot-tracked input
02	Serialised input
03	Quantity consumption
04	Mixed BOM genealogy

What to do

- 1 Use lot or batch information to trace bulk input material back to its source.
- 2 Use serialised Component tracking when each individual part must be bound to a finished unit.
- 3 Use quantity consumption for non-serialised Components.
- 4 Combine both tracking types in one BOM when the Product uses mixed materials.

Rules to remember

- Normal Fynd ERP Work Orders still create one finished-unit serial and Production Task per output unit.
- A serial number batch groups finished-unit serials; it is not a quantity-only production mode.
- Material genealogy is strongest when lots, serials, and consumption events are captured at the correct Station.

Ready when

- [] Every Component has the correct tracking type.
- [] Lot identifiers are available for batch-tracked inputs.
- [] The team understands that finished output remains unit-serialised.

STAGE 04

Design the Production Process

Turn the product definition into an executable route, connect Stations, assign Lines, and prepare identifiers.

5 guided articles

Routing in Plain Language

Understand how a Routing tells Fynd ERP where a unit goes and what happens at each operation.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/design-process/routing-basics>

Before you begin

- Activate a BOM Version
- Create Stations and Repair Stations

How it works

01	Start
02	Operation Station
03	Optional quality or packaging step
04	End

What to do

- 1 A Routing is created for a Product, one or more Variants, and an active BOM Version.
- 2 Each operation node points to a real Station and includes a mandatory Repair Station.
- 3 Connect Start, every operation stage, and End to define the execution order.
- 4 Conditional branches can use a quality result; lower priority numbers are evaluated first.

Rules to remember

- Assigned Components and quantities must both reach 100% BOM coverage.
- More than one Routing can be Active; the Routing is determined when the Work Order starts.
- Components and tools can be attached to the operation where they are needed.

Ready when

- [] I can explain the route from Start to End.
- [] Every operation maps to a real Station.
- [] Failure paths have an intentional destination.

Create and Validate a Routing

Build a connected production workflow that passes Fynd ERP's Routing validation.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/design-process/create-routing>

Before you begin

- Routing in Plain Language

How it works

01	Open Product
02	Select Variants
03	Select Active BOM
04	Add operation stage
05	Cover BOM quantities
06	Connect the workflow
07	Confirm and save

What to do

- 1 Open the Product detail and select Create Routing.
- 2 Select at least one Variant, continue, and select an Active BOM Version.
- 3 Enter the required Workflow Name, add an optional description, and set Active as needed.
- 4 Select Add Stage and Create Manually; choose the main Station, enter a unique Operation Name, and select a different Repair Station.
- 5 Assign the required Components and quantities until both BOM coverage indicators reach 100%.
- 6 Save the stage and connect Start to every stage and then End. Save becomes available when the workflow is valid.
- 7 Select Save, review the summary, select Confirm & Save, and verify the Routing detail page.

Rules to remember

- The Product is supplied by the Product detail page; at least one Variant, an active BOM Version, and a Workflow Name are required.
- A Routing needs at least one operation Station.
- Every operation requires a Repair Station different from its main Station.
- Component and quantity coverage must both reach 100% of the selected BOM.
- Start, every operation stage, and End must be connected; Save remains unavailable until automatic validation passes.

Ready when

- [] The intended Product Variants and Active BOM are selected.
- [] Every operation has valid main and Repair Stations and full BOM coverage.
- [] Start, every stage, and End are connected and the workflow is valid.
- [] The Routing detail page shows the expected status, Product, Variants, BOM, and Workflow Preview.

Assign Lines to Product

Tell Fynd ERP which active Lines are allowed to receive tasks for a Product.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/design-process/product-line-assignment>

Before you begin

- Create Production Lines
- Create Products and Variants

How it works

01	Open Product
02	Open Assigned Lines
03	Review Line availability
04	Assign eligible Line
05	Verify assignment

What to do

- 1 Open the Product detail and scroll to Assigned Lines.
- 2 Select Assign Line and open the Select Line control.
- 3 Review Available Lines separately from Lines assigned to other Products; use the code and Site to choose the correct Line.
- 4 Select Assign, then repeat only for other Lines that can physically manufacture the same Product.
- 5 Verify each assigned Line's name, code, status, capacity per hour, Site, and actions in the table.

Rules to remember

- At least one Line assignment is needed to enable multi-line Work Order execution for the Product.
- Assign only Lines that can physically manufacture the Product.
- Starting-Line eligibility is configured separately on the Line, not in the Product assignment dialog.

Ready when

- [] The Product has at least one eligible assigned Line.
- [] Assigned Line codes and Sites match the intended physical process.
- [] The table shows the expected status and capacity for every assignment.

Starting Lines and Capacity

Configure where newly released units enter production and how Fynd ERP prevents Line overload.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/design-process/starting-lines-and-capacity>

Before you begin

- Assign Lines to Product

How it works

01	Open Line detail
02	Edit throughput
03	Limit task queue
04	Set Starting Line
05	Save and verify

What to do

- Open Process & Engineering -> Line, search by name or code, open the Line detail, and select Edit.
- Optionally set Capacity Per Hour and Capacity Per Shift; when both are supplied, the shift value must be greater than the hourly value.
- Optionally set Task Queue Capacity to the maximum pending and active tasks allowed on the Line.
- Turn on Is Starting Line only when the Line may receive the initial task during capacity planning.
- Select Save and verify the status, Site, capacity values, and assigned Products on the Line detail.

Rules to remember

- Only Lines marked as Starting Lines can receive the initial task during capacity planning.
- The UI identifies the manufacturing phase order as SMT -> BLT -> ASSEMBLY -> TESTING -> PACKING.
- Capacity Per Shift must be greater than Capacity Per Hour when both optional values are supplied.
- Task Queue Capacity is optional; when supplied, it must be a whole number from 1 to 100,000.

Ready when

- [] Starting-Line status reflects the real first manufacturing phase.
- [] Hourly, shift, and queue capacity values are valid and realistic.
- [] The Line detail shows the intended Site, status, capacities, and assigned Products.

Configure Product Identifiers

Prepare serial, range, template, or bundle identifiers that must be assigned when Production Tasks are created.

CONDITIONAL

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/design-process/product-identifiers>

Before you begin

- Create Products and Variants

How it works

01	Create identifier definition
02	Choose source and type
03	Define identifier pool
04	Open Product Identifiers
05	Configure Product mapping
06	Set assignment behavior
07	Verify mapping

What to do

- 1 Open Process & Engineering -> Identifier Management and select Create Identifier Range -> Create Manually.
- 2 Enter the required Name and choose Range, Template, External, or Manual as the source.
- 3 For a Range source, select the identifier Type and Range or File Upload. Enter required Start and End values for a generated range, review its count, and select Create.
- 4 Open Production -> Products, open the Product detail, scroll to Product Identifiers, and select Add Identifier.
- 5 Select the identifier definition, enter the required Identifier Key and Display Name, and choose Generated or External.
- 6 Choose On Task Creation, On Task Start, or At Route Step; enable Auto-generate and Must be Unique when required.
- 7 Select Add Identifier and verify the Product table shows the expected source, timing, and linked range.

Rules to remember

- Identifier range Name is required, and a single operation can process no more than 10,000 identifiers.
- Generated identifiers are created by the system; External identifiers are scanned or entered by an operator.
- Product Identifier Keys are stored in uppercase.
- At Route Step timing requires configuration in the Routing Editor.

Ready when

- [] The identifier definition is Active and exposes the intended values and availability.
- [] The Product mapping has the correct key, type, source, and assignment timing.
- [] Auto-generation and uniqueness match the identifier's business rule.

STAGE 05

Plan and Launch Production

Create production demand, generate unit serials, plan Line capacity, validate readiness, and release work safely.

5 guided articles

Production Order vs Work Order

Choose the right planning record and understand which one creates executable production.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/plan-production/production-order-vs-work-order>

Before you begin

- Create and Validate a Routing
- Assign Lines to Product
- Create Shifts

How it works

01	Optional Production Order
02	One or more Work Orders
03	Production Tasks

What to do

- 1 Use a Production Order to group and monitor related Work Orders. Name and a unique Order Number are required; Client is optional and restricts the Work Orders that can be grouped.
- 2 Use a Work Order as the executable manufacturing record for a Product with Routing, input quantity, priority, due date, and Shifts.
- 3 Use Test Order only for production-line setup or workflow testing because Test Orders are excluded from analytics.

Rules to remember

- A Production Order groups Work Orders and does not define executable quantity by itself.
- A Work Order can exist without being grouped under a Production Order.
- At least one Shift is required before a Work Order can be started.
- Test Orders are not calculated in production analytics.

Ready when

- [] Grouping demand uses a Production Order with the intended client restriction.
- [] Every executable quantity is represented by a routed Work Order.
- [] Test Order is enabled only for non-analytics setup or workflow testing.

Create a Work Order

Create an executable production instruction after all Product and factory dependencies are ready.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/plan-production/create-work-order>

Before you begin

- Production Order vs Work Order
- Create and Validate a Routing
- Assign Lines to Product
- Create Shifts

How it works

01	Enter required name
02	Select routed Product and Variant
03	Enter quantity, priority, and due date
04	Assign Shift when known
05	Create Pending Work Order

What to do

- 1 Enter a unique Work Order name.
- 2 Select a Product with a configured Routing, then select the required Product Variant; Project and Product Family are derived automatically.
- 3 Enter a positive Input Quantity, choose the required Priority, and select a Due Date that is not in the past.
- 4 Review Total Computed Quantity, which equals Input Quantity multiplied by the Product Multiplier.
- 5 Assign one or more Shifts when known, review Component Availability, and create the Work Order.

Rules to remember

- Name, routed Product, Product Variant, Input Quantity, Priority, and Due Date are required.
- The Due Date cannot be in the past, and computed quantity equals Input Quantity x Product Multiplier.
- Shift is optional at creation, but at least one Shift is required before the Work Order can start.
- Test Orders are intended for line setup or workflow testing and are excluded from analytics.
- A successfully created Work Order begins in PENDING status.

Ready when

- [] Product, Variant, quantity, priority, and due date are correct.
- [] The computed quantity and Component Availability panel are reviewed.
- [] A Shift is assigned now or planned before start, and Test Order is off for normal production.

Generate and Confirm Unit Serials

Create exactly one finished-unit serial for every unit before releasing a normal Work Order.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/plan-production/generate-unit-serials>

Before you begin

- Create a Work Order

How it works

01	Open start journey
02	Choose generation method
03	Preview and validate uniqueness
04	Generate and confirm batch
05	Verify confirmed quantity

What to do

- 1 Select Start Work Order; the Serial Numbers step blocks progress until confirmed serials equal the Work Order quantity.
- 2 Select Generate Serial Numbers and use the Product Template when configured, another available Template, or Manual Upload for CSV or Excel.
- 3 Verify the preview count, select Validate Uniqueness, and generate the batch.
- 4 Select Confirm & Complete, then verify CONFIRMED and the exact serial count in the start journey.

Rules to remember

- Confirmed serial count must exactly equal the Work Order quantity before the next start step is enabled.
- Product Template is unavailable when no template is configured for the Product.
- Every proposed serial must pass uniqueness validation.
- A GENERATED batch is not production-ready until Confirm & Complete changes it to CONFIRMED.
- A serial number batch groups unit serials; it does not replace unit-level identities.

Ready when

- [] The serial batch status is CONFIRMED.
- [] The confirmed serial count equals Work Order quantity.
- [] Serial format is suitable for scanning and traceability.

Plan Capacity and Select Routing

Confirm eligible Starting Lines have enough headroom and choose the active Product Routing.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/plan-production/capacity-and-routing>

Before you begin

- Generate and Confirm Unit Serials
- Starting Lines and Capacity

How it works

01	Open Capacity and Lines
02	Review eligible Lines
03	Plan task assignments
04	Select active Routing
05	Review blockers

What to do

- 1 After confirmed serials match quantity, open Capacity & Lines and select Configure Capacity Planning.
- 2 Review Total Quantity, Available Capacity, Available Lines, and projected capacity on eligible Starting Lines.
- 3 Use system-optimized distribution or manual assignment; in system mode, select Proceed with System Assignment.
- 4 Confirm capacity planning completed, then select the eligible active Production Route.
- 5 Verify the route option's BOM status, route type, item count, and enabled Next action.

Rules to remember

- Capacity planning is disabled until serials are confirmed for the full Work Order quantity.
- Available queue capacity on eligible Starting Lines must cover the Work Order quantity.
- The selected Production Route must be eligible and show an active associated BOM.

Ready when

- [] Capacity planning is complete.
- [] No Line exceeds its queue capacity.
- [] The correct active Routing is selected.

Pre-Start Validation and Release

Use Fynd ERP's readiness checks to fix blockers before Production Tasks are created.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/plan-production/pre-start-validation>

Before you begin

- Plan Capacity and Select Routing

How it works

01	Run validation
02	Resolve blockers
03	Transfer first-stage material
04	Review warnings
05	Release

What to do

- 1 After Serial Numbers, Capacity & Lines, and Routing are complete, select Run Pre-Start Validation.
- 2 Review Product assignment, BOM, generated and confirmed serials, Component Inventory, Station-Level Inventory, and Identifier Availability.
- 3 Resolve first-stage Station shortages with Transfer All or a component transfer; use Auto-Fill Shortage for the exact remaining quantity.
- 4 Refresh until validation shows PASS, then select Proceed to Start.

Rules to remember

- A first-stage Station shortage blocks start even when plant-level Component Inventory is sufficient.
- Transfer All prioritizes storage locations, then non-routing Stations, and may perform partial transfers.
- Proceed to Start releases immediately, creates Production Tasks, and changes the Work Order to IN PROGRESS.
- Validation must show PASS before Proceed to Start is enabled.

Ready when

- [] All blockers are resolved.
- [] Warnings have an understood mitigation.
- [] The Work Order changes to In Progress and creates the expected number of Production Tasks.

STAGE 06

Run Production

Understand Production Tasks, operator execution, material consumption, and the status trail created during work.

4 guided articles

Production Tasks

Understand the unit-level record operators execute after a Work Order is released.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/run-production/production-tasks>

Before you begin

- Pre-Start Validation and Release

How it works

01	Create one task per serial
02	Assign initial Line
03	Follow Routing
04	Record status and Route Logs

What to do

- 1 Open Production Tasks on the released Work Order and verify one Task exists for each confirmed serial.
- 2 Review task distribution by Line Type and Line, then open a unit Task.
- 3 Confirm Task status, assigned serial, Work Order, Product, Line, and Line Type.
- 4 Review Stage Progress, Bound Identifiers, BOM Progress, and Routing and Process Details.

Rules to remember

- Release creates one Production Task per confirmed finished-unit serial.
- Capacity planning provides the initial Line assignment, and the selected Routing defines the current Stage and Station.
- Required BOM components remain Pending until consumed during execution.
- Generated Product identifiers can be bound to the Task before execution begins.

Ready when

- [] Task count matches Work Order quantity.
- [] Every Task has a serial and eligible Line.
- [] Operators can find Tasks at the correct Station.

Operator Execution Flow

Follow the normal sequence an operator uses to process a unit at a Station.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/run-production/operator-flow>

Before you begin

- Production Tasks

How it works

01	Select mapped Station
02	Choose Work Order and Line
03	Open Production Task
04	Validate Component identities
05	Complete and record

What to do

- 1 In Operator, select the Product first, choose Mapped, and use the Status filter if the Station is hidden by the default Running view.
- 2 Open the Station, select an active Work Order, and choose an eligible Production Line.
- 3 Scan or enter the Production Task ID or confirmed finished-unit serial, then select Start Task.
- 4 Enter an available serial for every serialised BOM Component and satisfy each configured identifier validator.
- 5 Confirm the result and select Complete & Next so Fynd ERP records execution and Route progress.

Rules to remember

- Mapped Stations are product-aware; select the Product before using the Mapped filter.
- Only active Work Orders and eligible Production Lines are offered for the selected Station.
- Required Component identities must be available and valid; the Battery Pack validator requires GS1 AI 01 and AI 10.
- Complete & Next records the operator, Station, material identities, result, timestamps, and Route progress.

Ready when

- [] The Product, Station, Work Order, Line, Task, and Routing step match.
- [] Every required Component identity validates successfully.
- [] The completion is visible in Task History and Route history.

Material Consumption During Production

Capture the material used by each unit so inventory and genealogy remain accurate.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/run-production/material-consumption>

Before you begin

- Operator Execution Flow
- Track Serial, Lot, and Mixed Materials in Item Master

How it works

01	Transfer material to Station
02	Open Production Task
03	Validate Component identities
04	Complete consumption
05	Verify genealogy

What to do

- 1 Transfer required stock to the execution Station and confirm the Station inventory covers the BOM requirement.
- 2 Scan or enter the Production Task ID or confirmed finished-unit serial in the Station work area.
- 3 Enter one available, unconsumed serial for every serialised BOM Component and satisfy its configured identifier validator.
- 4 Select Complete & Next after all required inputs validate.
- 5 Verify the Production Task is COMPLETED and the material identities appear in consumption and genealogy history.

Rules to remember

- Serialised Components require a unique available serial; an already consumed serial is rejected.
- Configured identifier validators apply during execution; the Battery Pack requires GS1 AI 01 and AI 10.
- Material should be transferred before the unit reaches the Station that consumes it.
- Completion records material identities, inventory use, and genealogy against the Production Task.

Ready when

- [] Station inventory covers current production.
- [] Every serialised Component identity is unique, available, and valid.
- [] The completed Task and consumed identities are visible in production and traceability history.

Statuses and Route Logs

Read the history Fynd ERP creates as Work Orders and units move through production.

RECOMMENDED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/run-production/statuses-and-route-logs>

Before you begin

- Production Tasks

How it works

01	Pending
02	Starting
03	In Progress
04	Completed or exception state

What to do

- 1 Open the Work Order to review overall status, progress, Production Metrics, and task distribution.
- 2 Open Route Execution Logs and compare Total Logs, Passed, Failed, and In Progress.
- 3 Filter by status, date range, Work Order, Product, Routing, or Station.
- 4 Review the timestamp, production context, operator, quality result, and execution status in each log.

Rules to remember

- Work Order status and Task status answer different questions.
- A Work Order becomes COMPLETED only after every Production Task finishes.
- Route Execution Logs preserve operation-level evidence, including the Station, operator, quality result, status, and timestamp.

Ready when

- [] Work Order progress and task distribution agree.
- [] Completed operations appear in Route Execution Logs.
- [] Filters return the expected Work Order, Product, Routing, or Station history.

STAGE 07

Quality and Exceptions

Configure inspections and handle failures through defects, NCR, repair, rework, hold, scrap, and teardown.

4 guided articles

Configure Quality Controls

Decide where inspection, sampling, and verification belong in the manufacturing route.

CONDITIONAL

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/quality-exceptions/quality-controls>

Before you begin

- Create and Validate a Routing

How it works

01	Identify quality gate
02	Create Control Point
03	Bind scope
04	Define outcomes

What to do

- 1 Open Quality -> Control Points and select Create Control Point.
- 2 Enter the unique Code and Name, then select Control Type, Frequency, and On Failure.
- 3 Select the Production Line first, then choose the Station that becomes available.
- 4 Choose the failure action, enable Auto-hold failed units when required, and add checklist questions.
- 5 Create the control and verify its Active status, scope, failure action, and checklist on Configuration.

Rules to remember

- Code, Name, Control Type, Frequency, and On Failure are required.
- Station remains unavailable until a Production Line is selected.
- Failure action can Stop Line, Quarantine, Hold Lot, Escalate, or Alert Only.
- Checklist questions define the evidence operators must record.

Ready when

- [] The Control Point is Active and has a unique Code.
- [] Line and Station scope are correct.
- [] Failure action, Auto-hold behavior, and checklist questions match the production risk.

Inspections, Defects, NCR, and CAPA

Run a configured inspection, record failed samples, and confirm whether NCR and CAPA actions are enabled for the current configuration and role.

CONDITIONAL

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/quality-exceptions/inspection-to-capa>

Before you begin

- Configure Quality Controls

How it works

01	Active inspection configuration
02	Work Order or Lot inspection
03	Sample checks and Defect
04	Rejected inspection and hold scope
05	Confirm NCR and CAPA availability

What to do

- 1 Create and activate the applicable inspection configuration before starting an inspection.
- 2 Choose either a Work Order or an available, non-expired Lot-never both-and enter the required total and sample quantities.
- 3 Record each sample and its checks. A failed sample requires a Defect code and Severity before it can be saved.
- 4 Reject the inspection and choose whether to hold only failed serials or the entire lot. Inspection-local defects and plant Defects are separate records.
- 5 Confirm NCR and CAPA availability for the current configuration and signed-in role. If a working create action is not present, confirm enablement with the implementation owner before using the workflow.

Rules to remember

- An active inspection configuration is a prerequisite; Plant scope is the fallback and more specific scopes override it.
- A source is either a Work Order or a Lot, and total quantity is required.
- A failed sample cannot be saved without Defect details.

Ready when

- [] The inspection configuration is Active.
- [] The source, quantities, sample serial, checks, and Defect details are complete.
- [] The selected hold scope matches the rejected output.

Repair and Rework

Configure repair intake, record rework evidence, and verify the resulting repair or re-inspection state.

CONDITIONAL

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/quality-exceptions/repair-and-rework>

Before you begin

- Create Stations and Repair Stations
- Create and Validate a Routing

How it works

01	Repair configuration
02	Debug Queue
03	Rework Board
04	Record work
05	Verify the result

What to do

- 1 In Repair Config, enable Auto-create repair on failure only when automatic intake is required. Set a Default Repair Station when automatically created tickets must be routed to that Station.
- 2 Use Debug Queue to triage incoming failures. Rework Board jobs originate from a failure or Disposition; the board has no create action.
- 3 Review the linked Work Order, serial, mechanism, attempt number, and current board column.
- 4 Select Record work and enter labor minutes, material cost, and concise work notes before selecting Complete Work.
- 5 After Complete Work, refresh the job and confirm that its status, notes, labor, costs, timeline, and any required re-inspection state were recorded before treating the handoff as successful.

Rules to remember

- Auto-create repair controls automatic queue intake; Default Repair Station controls station routing when that routing is required.
- Attempt Limit rules are configured under Alerts and can alert, block, or block and propose scrap.
- Total and per-line repair WIP caps are optional; leaving a cap empty disables that limit.

Ready when

- [] The job remains linked to the original serial and Work Order.
- [] Labor minutes, material cost, and work notes are entered.
- [] The completed work and any required re-inspection state are visible before the handoff is accepted.

Hold, Scrap, and Teardown

Create scoped holds, track controlled scrap dispatches, and start teardown only from eligible executed scrap.

CONDITIONAL

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/quality-exceptions/hold-scrap-teardown>

Before you begin

- Statuses and Route Logs

How it works

01	Choose Hold Level
02	Select target and Reason
03	Approval when required
04	Track Scrap dispatch
05	Send executed Scrap to Teardown

What to do

- 1 Open Quality -> Hold Management -> Create Hold and choose the Hold Level before selecting a target.
- 2 Select the required target and Reason. Notes are optional, and a Work Order hold can include an optional expected release date.
- 3 Submit inventory lot, vendor, and date-code holds for approval; they remain inactive until Quality approves them.
- 4 Configure Scrap Reasons first, then use Scrap Register to track Pending Approval, Approved, Executed, and Rejected dispatches. The register has no create action.
- 5 Use Send to teardown on an executed Scrap dispatch to create a Teardown order linked to the donor serial and recovered Components.

Rules to remember

- Hold Level determines the target and impact; a Work Order hold pauses all production for that order.
- Inventory lot, vendor, and date-code holds require Quality approval.
- Teardown cannot begin until Scrap is executed and explicitly sent to teardown.

Ready when

- [] The target and Reason are selected for the correct Hold Level.
- [] Scrap reason categories and reason codes exist before dispatch.
- [] The teardown order remains linked to its donor serial and Scrap dispatch.

STAGE 08

Package, Ship, and Trace

Configure container operations, verify shipment readiness, and trace finished units and materials across their lifecycle.

4 guided articles

Packaging in Fynd ERP

Understand how unit, container, label, weight, and closure operations fit into a Routing.

CONDITIONAL

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/package-trace/packaging>

Before you begin

- Create and Validate a Routing

How it works

01	Open Packaging
02	Configure product container hierarchy
03	Create a work-order container
04	Review the generated UID
05	Complete packaging checks

What to do

- 1 Open Packaging & Shipping and select Packaging.
- 2 Open Packing Configuration, select the Product, and define its product-specific container type. Product, Name, and Code are required; Code accepts 2-10 alphanumeric characters.
- 3 Leave Child Container Type empty for a leaf that binds finished-unit serials directly, or select a child to build the Device -> Box -> Carton -> Pallet hierarchy.
- 4 Return to the Packaging Dashboard, select Create Container, and choose the Product, required Container Type, and required Work Order ID.
- 5 Open the generated UID to review device count, weight and dimensions, linked Work Order, label state, and History.

Rules to remember

- Packing configurations are product-specific.
- A leaf has no child container type and binds serial numbers directly; parent configurations accept the selected child type.
- Create Container remains disabled until Container Type and Work Order ID are selected.
- A UID template is optional; when omitted, Fynd ERP uses the default UID format.
- Pallets are added to Shipments, which are managed separately in the Shipments module.

Ready when

- [] Product-specific container configuration is active.
- [] Device -> Box -> Carton -> Pallet hierarchy matches the physical flow.
- [] Container is linked to the correct Work Order.
- [] UID, device count, label state, and History are visible before closure.

Containers and Labels

Create a clear packaging identity from individual units through cartons and pallets.

CONDITIONAL

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/package-trace/containers-and-labels>

Before you begin

- Packaging in Fynd ERP

How it works

01	Configure leaf and parent types
02	Set UID and physical limits
03	Create the parent container
04	Pack an eligible identity
05	Print and audit the label

What to do

- 1 In Packing Configuration, create the leaf first, then create each parent with its allowed Child Container Type; selecting a child makes Max child count required.
- 2 Choose a container-specific UID Template and sequence reset scope, then configure dimensions, weight range, and tolerance.
- 3 Create the parent from the Packaging Dashboard by selecting the Product, parent Container Type, and completed Work Order.
- 4 In Pack by UID, scan a finished serial, device UID, child UID, identifier, or task ID; a valid unit automatically creates the configured leaf and packs it into the parent.
- 5 Print the carton label and verify carton UID, model, quantity, unit serial, QR codes, History, and the Reprint path.

Rules to remember

- Use a container-specific UID pattern so generated container identities do not collide with unit serials.
- A parent accepts only its configured child type, and its maximum child count is enforced.
- Only eligible production identities can be packed; source values unavailable to the label remain blank.

Ready when

- [] Leaf and parent type mirror the physical hierarchy.
- [] UID template, capacity, dimensions, weight, and tolerance are correct.
- [] Packing, printed label content, History, and Reprint have been checked.

Shipment Verification

Verify that packaged output matches shipment requirements before dispatch.

CONDITIONAL

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/package-trace/shipment-verification>

Before you begin

- Containers and Labels

How it works

01	Select the Shipment first
02	Check the active Shipment Rule
03	Confirm Shipping Gate balance
04	Configure ASN export
05	Verify deployment prerequisites

What to do

- 1 Open Shipment Verification and select an existing Shipment; Container Type, counts, container rows, and Export remain unavailable until the required Shipment is selected.
- 2 Confirm an active Shipment Rule includes the selected container configuration and permits its container-count and maximum-weight limits.
- 3 Confirm the Shipping Gate has positive remaining approved Purchase Order quantity.
- 4 Use a client- or product-scoped ASN template for the Excel export, and confirm that the required template is configured.
- 5 With a valid Shipment selected, verify shipment creation, rule matching, gate balance, container verification, and ASN output before operational use.

Rules to remember

- Shipment selection is mandatory before Container Type or verification data becomes available.
- The governing Shipment Rule and a positive Shipping Gate balance are prerequisites.
- ASN output is client- or product-scoped when a template is configured.

Ready when

- [] A valid Shipment exists and can be selected.
- [] The active rule covers the container configuration, count, and weight.
- [] Shipping Gate balance is positive and the required ASN template exists.

Traceability, Genealogy, and Recall

Follow a finished unit back to its Work Order, operations, Components, material lots, containers, and shipment.

RECOMMENDED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/package-trace/traceability-genealogy-recall>

Before you begin

- Material Consumption During Production
- Shipment Verification

How it works

01	Set dossier scope
02	Choose sheets
03	Export Stage Timeline
04	Search genealogy
05	Find affected units
06	Create Recall Notice

What to do

- 1 In Master Traceability, combine filters and unlock dependent Product filters from their parent selection. A specific identifier list overrides the other filters, and Generate Dossier remains disabled until at least one scope is selected.
- 2 Select only the required dossier sheets. Generation is asynchronous; confirm that Recent exports shows the expected unit count when the dossier completes.
- 3 For Stage Timeline, select a date range, at least one Work Order, and 1-50 Operations. Treat the date range as required even when the form does not mark it with an asterisk.
- 4 Use the full genealogy tree with a pallet, carton, box, device, product, Component, or batch identifier. Downstream follows a finished identity to Components and lots; Upstream follows a lot toward finished units.
- 5 Use Find Affected Units with a Batch/Lot Number or Component Serial, then confirm that the results contain the expected finished-unit serials.
- 6 Recall creation requires Title and Severity. Confirm that the selected batch or tasks populate affected units and that the saved notice records them before relying on the recall response.

Rules to remember

- Traceability quality depends on accurate scanning and consumption during production.
- A specific identifier list overrides the other dossier scope filters.
- Stage Timeline requires a date range, at least one Work Order, and 1-50 Operations.
- A recall notice controls the response; traceability identifies the affected scope, and the saved notice must show the assigned affected units.

Ready when

- [] The completed dossier appears in Recent exports with the expected unit count.
- [] The selected finished-unit serial resolves to its consumed Components and material lots.
- [] The selected material lot returns the expected affected finished units.
- [] The saved Recall Notice contains the intended affected units and status.

STAGE 09

Operate and Improve

Use dashboards, shift handovers, maintenance, tools, reports, and audit history to keep production controlled.

4 guided articles

Dashboards and Production Monitoring

Choose the dashboard that answers the operational question instead of treating every metric as the same.

RECOMMENDED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/operate-improve/dashboards>

Before you begin

- Statuses and Route Logs

How it works

01	Plant dashboard
02	Work Order risk
03	Work Order metrics
04	Live progress board
05	Production Task status
06	Route Log evidence

What to do

- 1 Use the Dashboard to review setup readiness, active Work Orders, production Stations, active Products, Clients, Projects, production completion, and quality targets for the selected Plant.
- 2 In Work Orders, evaluate Status, Priority, and Due Date together. Past-due active records are marked Delayed; use Work Order identity and context to distinguish original production from later repair work.
- 3 Open a completed Work Order to review completed tasks, units per hour, takt time, cycle time, yield, and total execution time. Yield is passed quality steps divided by completed steps; compare it with the displayed target.
- 4 Use Track Progress for the live board. It auto-refreshes every 30 seconds by default and supports Station filters, completed highlighting, samples-only, sorting, and fullscreen.
- 5 Use Production Tasks to verify Serial, task ID, grouping, Work Order, Line, Product/Routing, and status. A serial can have multiple task records, including a later repair task, so confirm the Work Order and Line before acting.
- 6 Use Route Logs to verify the Work Order, Product, Routing, Line, Station, operator, timestamp, quality status, and execution status. Confirm that the filtered logs match the completed work.

Rules to remember

- A dashboard summarises records; the underlying records remain the source of evidence.
- Route Log summary totals reflect the current filters.
- Takt and cycle metrics use completed execution time; yield is passed quality steps divided by completed steps.
- The same serial can have multiple task records, so task status must be interpreted with its Work Order and Line.

Ready when

- [] Plant-level counts and targets match the selected Plant.
- [] Original production and later repair Work Orders are distinguished by identity, context, and status.
- [] The live board shows the expected completed tasks and execution logs.
- [] Filtered Route Logs show the expected quality and execution status with operator and Station evidence.

Shift Operations and Handovers

Maintain continuity when production responsibility moves from one Shift to another.

RECOMMENDED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/operate-improve/shift-operations>

Before you begin

- Create Shifts
- Dashboards and Production Monitoring

How it works

01	Review Shift Definitions
02	Verify Schedule coverage
03	Create Handover
04	Record connected conditions
05	Review Break Compliance

What to do

- 1 Confirm net working time after unpaid breaks. For example, a 06:00-14:00 Shift with a 15-minute unpaid break has 7h 45m net working time.
- 2 Verify Shift Schedule coverage. New Schedules remain Draft until activated, but status alone does not prove coverage; confirm generated shifts per week and calendar coverage.
- 3 Create a Shift Handover. Shift and Handover Date are required; Line, outgoing/incoming operators, summaries, notes, pending tasks, and checklist items are optional.
- 4 Connect the Handover to the correct Shift and Line, then record completed production, Route Log results, equipment state, safety, and pending repair tasks.
- 5 Review the Draft before completion. Confirm the Outgoing and Incoming Supervisors explicitly, and use Complete only after the next Shift reviews the record.
- 6 Use Break Compliance carefully: a 100% result with zero Break Logs is not evidence that compliant breaks occurred. Confirm the underlying logs; Export CSV requires exportable records.

Rules to remember

- Unpaid breaks are deducted from Shift duration when net working time is calculated.
- A Schedule marked Active can still have zero assigned shifts and no effective coverage.
- A new Handover is Draft; Complete it only after the receiving Shift reviews the recorded context.
- A 100% Break Compliance rate with zero Break Logs does not prove compliant breaks occurred.

Ready when

- [] Shift net working time matches the schedule after unpaid breaks are deducted.
- [] The Handover is connected to the correct Shift and Line.
- [] Production, quality, equipment, safety, and pending repair work are recorded.
- [] The Draft is reviewed before Complete is selected.

Tools, Equipment, and Maintenance

Keep required production assets available, correctly located, and safe to use.

CONDITIONAL

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/operate-improve/tools-and-maintenance>

Before you begin

- Create Stations and Repair Stations

How it works

01	Register the Tool
02	Set calibration governance
03	Record calibration evidence
04	Evaluate Maintenance Alerts
05	Approve the Maintenance Work Order

What to do

- 1 Open the child Tools and Equipments page and supply the required Tool Name, Tool Code, Tool Category, Manufacturer, Model, Station, and Status.
- 2 Enable Requires Calibration only when the Tool is governed; this reveals interval and provider fields.
- 3 Record Calibration Date, Next Calibration Date, and Result so a governed Tool moves from Unknown to a current calibration state.
- 4 Use Trigger Maintenance Check to evaluate configured rules. A result of 0 alerts may mean that no rule matched or that required rules or source data are absent; verify both before relying on the result. Export requires alert rows.
- 5 Create a Maintenance Work Order with the required Tool and Title, then move it through Draft, Submitted, and Approved before Start is available.

Rules to remember

- Tool Name, Tool Code, Tool Category, Manufacturer, Model, Station, and Status are required.
- A calibration-governed Tool remains Unknown until a calibration record defines its result and next due date; overrides remain auditable.
- Maintenance Work Orders require Tool and Title, default to Medium and Corrective, and cannot be started before approval.

Ready when

- [] The Tool is assigned to the intended production Station and has the correct status.
- [] Calibration evidence, certificate, and next due date are current when required.
- [] The approved maintenance window is scheduled and the order has not been started early.

Reports, Exports, and Audit Trail

Use reports and immutable history to explain production results and configuration changes.

RECOMMENDED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/operate-improve/reports-and-audit>

Before you begin

- Dashboards and Production Monitoring

How it works

01	Set combined dossier scope
02	Choose report sheets
03	Generate and verify the XLSX
04	Create a filtered Work Order export
05	Review available change history

What to do

- 1 In Master Traceability, combine optional scope filters or use a pasted/uploaded serial-number list, which overrides the other scope filters.
- 2 Select the required dossier sheets, generate the report, and wait for Recent exports to show Completed, the unit count, and the XLSX filename.
- 3 From Work Orders, open Export Work Orders to use the separate Create Data Export dialog. Re-apply Search, Status, and Priority when list filters are not inherited.
- 4 Open Export Details to confirm type, format, state, file size, applied filters, processed rows, and progress before downloading.
- 5 Use View all changes and Recent Changes for the available actor, action, affected record, and timestamp evidence. Confirm that the chosen history view contains the required evidence before relying on it.

Rules to remember

- A serial-number list overrides the other Master Traceability scope filters.
- A Completed export is not sufficient evidence by itself; verify processed rows, progress, file size, applied filters, and file contents.
- The required audit evidence is available from the chosen history view.

Ready when

- [] The dossier scope and selected sheets match the investigation question.
- [] The operational export contains the intended filtered records.
- [] Recent Changes identifies who changed the selected Work Order and when.

STAGE 10

Help and Reference

Use checklists, requirement matrices, terminology, and decision trees when setting up or troubleshooting Fynd ERP.

5 guided articles

Required, Recommended, and Conditional Setup

Use one matrix to understand which records block production and which improve control.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/help-reference/requirement-matrix>

Before you begin

- Complete Fynd ERP Setup Journey

How it works

01	Required foundation
02	Recommended organisation
03	Conditional capabilities

What to do

- 1 Required: Site for Stations; Lines; operation and Repair Stations; Shifts; Components; Product and BOM; active BOM; Routing; Product-to-Line assignment; confirmed unit serials.
- 2 Recommended: Projects, Product Families, explicit capacity, Site assignment on Lines, dashboards, handovers, reporting, and trial production.
- 3 Conditional: Product Variants, additional identifiers, tools, automations, quality sampling, packaging, shipment, repair, rework, and recall configuration.

Rules to remember

- Required labels reflect the supported normal production journey.
- Recommended records may not be database blockers but prevent avoidable operational confusion.
- Conditional features become required when the selected production design depends on them.

Ready when

- [] Required setup is complete.
- [] Recommended structure has been reviewed.
- [] Every conditional capability has an explicit use or exclusion decision.

First Production Run Checklist

Run a small controlled Work Order to validate the complete configuration before scaling volume.

REQUIRED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/help-reference/first-production-run>

Before you begin

- Pre-Start Validation and Release

How it works

01	Create small Work Order
02	Release
03	Execute every operation
04	Test failure path
05	Package and trace
06	Review

What to do

- 1 Use a small quantity that is large enough to test serials, capacity, and material consumption.
- 2 Run at least one unit through the complete normal route.
- 3 Test a controlled failure and Repair path.
- 4 Verify quality, packaging, and shipment steps that apply.
- 5 Trace a completed unit back to material and forward to its container or shipment.

Rules to remember

- Do not use the first full-volume order as the first complete process test.
- Record every configuration issue and retest after correction.
- The trial is complete only when the unit history is understandable end to end.

Ready when

- [] Work Order creates the expected Production Tasks.
- [] Normal and failure paths work.
- [] Material consumption and inventory are correct.
- [] Packaging and shipment work where configured.
- [] End-to-end traceability is complete.

Troubleshooting Decision Tree

Start with the failing action and work backward through its dependencies.

RECOMMENDED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/help-reference/troubleshooting>

Before you begin

- How Fynd ERP Data Connects

How it works

01	Identify blocked action
02	Read exact message
03	Check immediate prerequisite
04	Check status and Plant
05	Correct
06	Retry

What to do

- 1 If Product creation fails, check BOM coverage and Variant rules.
- 2 If Routing creation fails, check active BOM, Stations, Repair Stations, and connectivity.
- 3 If Work Order creation fails, check Routing, active Line assignment, Shift, Product, and Variant.
- 4 If release fails, check serial count, Starting Line, capacity, active BOM, inventory, and required identifiers.
- 5 If an operation cannot complete, check current route, Station, required inputs, quality gates, and exception status.

Rules to remember

- Keep the exact error message; do not troubleshoot from memory.
- Confirm the record belongs to the current Plant.
- Fix the earliest missing dependency instead of changing downstream data.

Ready when

- [] The blocked action is clearly identified.
- [] The immediate prerequisite has been checked.
- [] The correction is verified with the smallest safe retry.

Fynd ERP Glossary

Use globally neutral definitions for the core records and actions in Fynd ERP.

RECOMMENDED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/help-reference/glossary>

How it works

01	Factory records
02	Product records
03	Process records
04	Execution records
05	Quality and trace records

What to do

- 1 Plant: the manufacturing context that owns and isolates operational data.
- 2 Site: a physical location within the Plant.
- 3 Line: a production lane that receives unit-level tasks.
- 4 Station: a work point used by a Routing operation.
- 5 BOM: the Components and quantities required for a Product.
- 6 Routing: the connected sequence of operations a unit follows.
- 7 Work Order: the executable instruction to produce a Product quantity.
- 8 Production Task: one finished unit moving through production.
- 9 Genealogy: the relationship between finished units, consumed material, packaging, and shipment.

Rules to remember

- Use Product for finished output and Component for input material.
- Use Production Order for grouping and Work Order for execution.
- Use Repair for fixing a failed unit and Rework for repeating controlled manufacturing operations.

Ready when

- [] Terminology is used consistently in procedures and training.
- [] Local factory terms are mapped to the Fynd ERP term.
- [] Users can distinguish similarly named records.

Capability Availability Guide

Recognise which capabilities are always part of the core flow and which appear only when configured.

RECOMMENDED

<https://forge-mes-cookbook-dcca2d61.serverless.boltic.app/cookbooks/ERP/Forge/help-reference/capability-availability>

Before you begin

- [Fynd ERP in Plain Language](#)

How it works

01	Core execution
02	Configured capability
03	Product-specific use
04	Plant-specific availability

What to do

- 1 Core flow includes factory records, Products, BOMs, Routings, Lines, Shifts, Work Orders, serials, Production Tasks, and history.
- 2 Quality, repair, packaging, identifiers, tools, automations, and shipment depend on configuration.
- 3 Some capabilities can be Product-specific, Variant-specific, Routing-specific, Station-specific, or Plant-specific.
- 4 Confirm availability in the current Fynd ERP environment before writing local procedures.

Rules to remember

- Do not assume every menu item is required for every manufacturing process.
- Do not describe proposed or unverified capabilities as available.
- Use the latest verified implementation and active configuration as the source of truth.

Ready when

- [] Core and conditional capabilities are clearly separated.
- [] Configured features have been tested in the current environment.
- [] Local training includes only available capabilities.